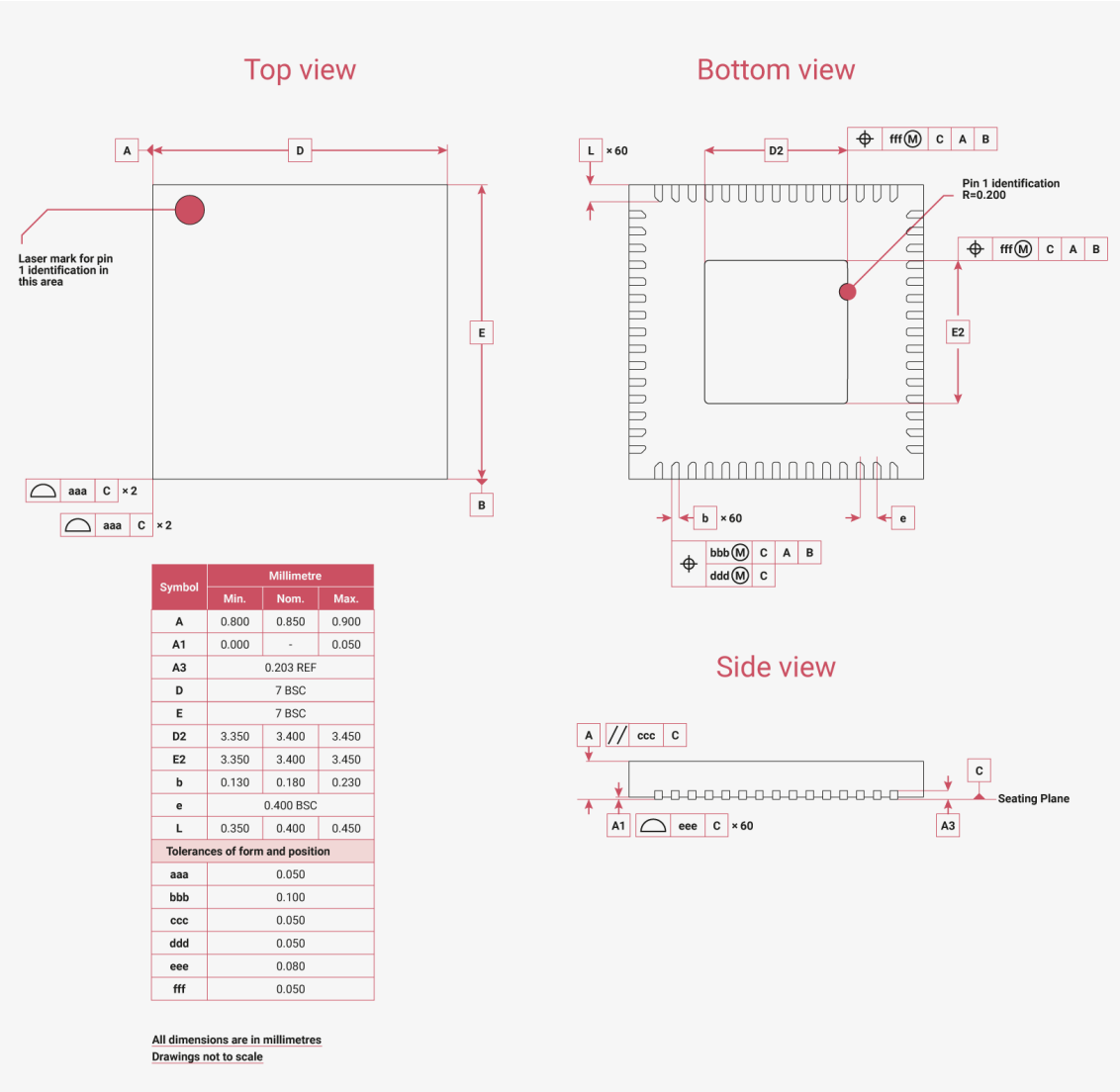


Chapter 14. Electrical and Mechanical

This section contains physical and electrical details for RP2350.

14.1. QFN-60 Package

Figure 142. Top down view (left, top) and side view (right, bottom), along with bottom view (right, top) of the RP2350 QFN-60 package



NOTE

Leads have a matte Tin (Sn) finish. Annealing is done post-plating, baking at 150°C for 1 hour. Minimum thickness for lead plating is 8 microns, and the intermediate layer material is CuFe2P (roughened Copper (Cu)).

14.1.1. Thermal characteristics

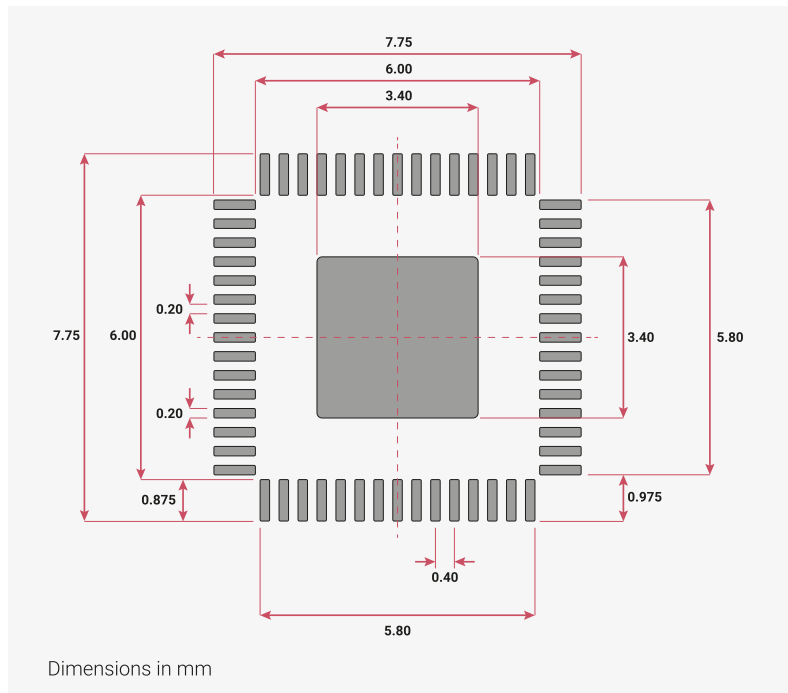
The thermal characteristics of the QFN-60 package are shown in [Table 1422](#).

Table 1422. Thermal data for the QFN-60 package.

Device	θ_{JA} (°C/W) - Still Air	θ_{JA} (°C/W) - 1m/s Forced Air	θ_{JA} (°C/W) - 2m/s Forced Air	θ_{JB} (°C/W)	θ_{JC} (°C/W)
RP2350A	40.542	31.99	30.264	12.588	14.315
RP2354A	TBD	TBD	TBD	TBD	TBD

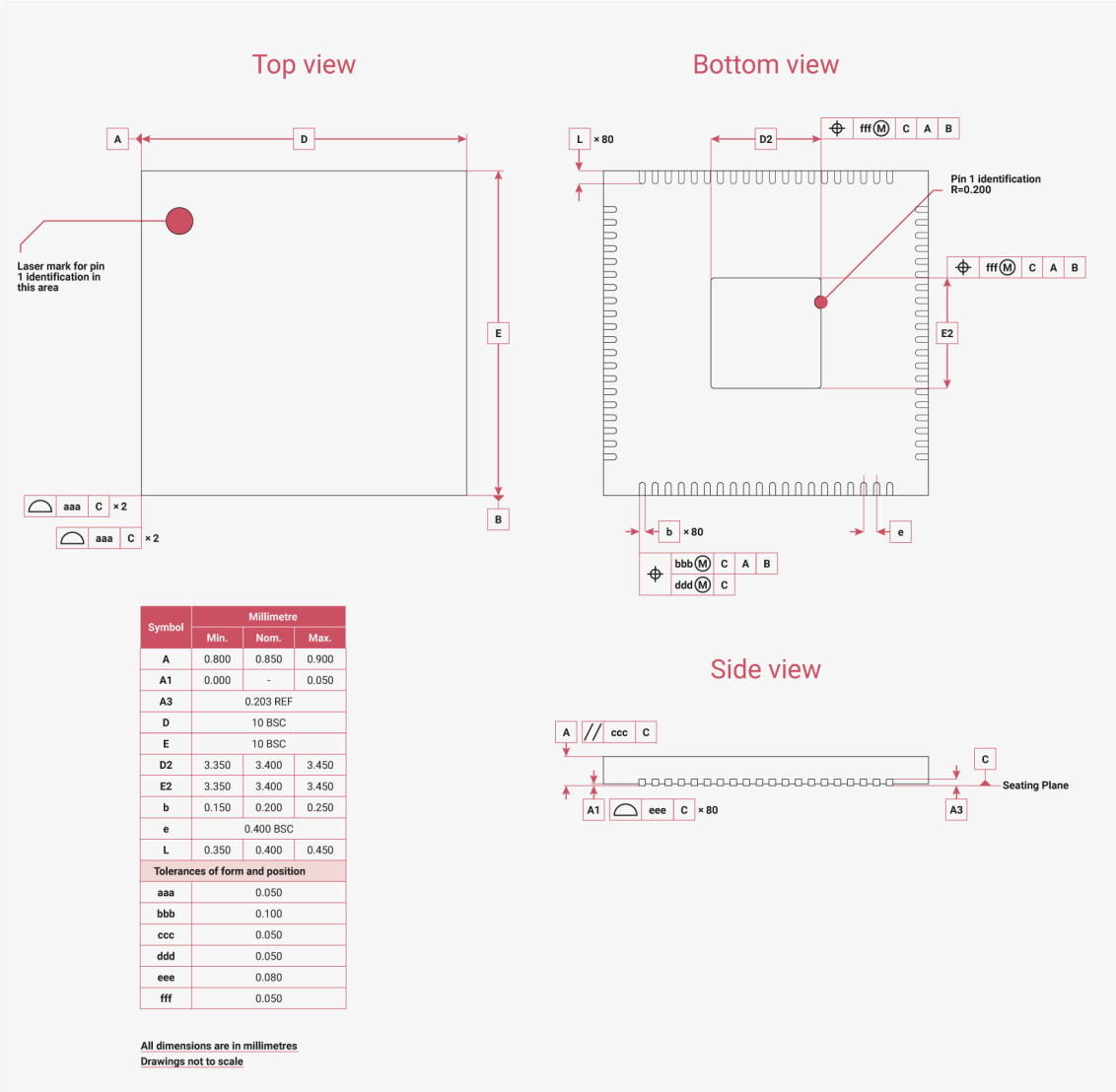
14.1.2. Recommended PCB Footprint

Figure 143.
Recommended PCB
Footprint for the
RP2350 QFN-60
package



14.2. QFN-80 Package

Figure 144. Top down view (left, top) and side view (right, bottom), along with bottom view (right, top) of the RP2350 QFN-80 package



NOTE

Leads have a matte Tin (Sn) finish. Annealing is done post-plating, baking at 150°C for 1 hour. Minimum thickness for lead plating is 8 microns, and the intermediate layer material is CuFe2P (roughened Copper (Cu)).

14.2.1. Thermal characteristics

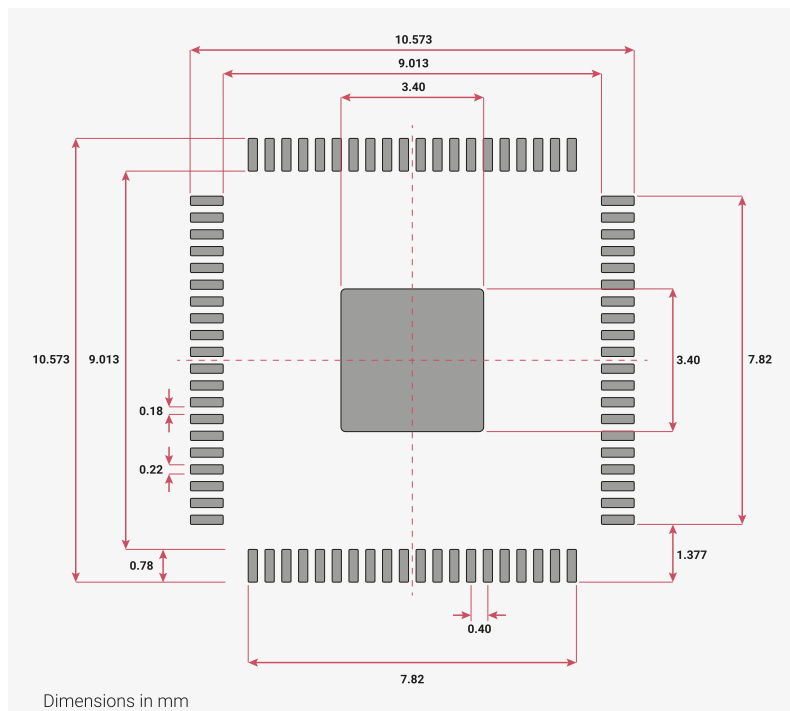
The thermal characteristics of the QFN-80 package are shown in [Table 1423](#).

Table 1423. Thermal data for the QFN-80 package.

Device	θ_{JA} (°C/W) - Still Air	θ_{JA} (°C/W) - 1m/s Forced Air	θ_{JA} (°C/W) - 2m/s Forced Air	θ_{JB} (°C/W)	θ_{JC} (°C/W)
RP2350B	TBD	TBD	TBD	TBD	TBD
RP2354B	TBD	TBD	TBD	TBD	TBD

14.2.2. Recommended PCB Footprint

Figure 145.
Recommended PCB
Footprint for the
RP2350 QFN-80
package



14.3. Flash in Package

RP2354A and RP2354B feature 2 MB of internal flash. In all other respects, including pinout, they are identical to their flashless counterparts RP2350A and RP2350B. They use the same QFN-60 (RP2354A) and QFN-80 (RP2354B) packages. An RP2354 device contains two stacked silicon die:

- the same RP2350 die as the flashless variants
- a Winbond W25Q16JVWI QSPI NOR flash

Winbond W25Q16JVWI Datasheet

For detailed information on the W25Q16JVWI device used in RP2354 see:

www.winbond.com/hq/product/code-storage-flash-memory/serial-nor-flash/?__locale=en&partNo=W25Q16JV

The six dedicated QSPI pads on the RP2350 die (**CSn**, **SCK** and **SD0** through **SD3**) connect to both the internal flash die and the external package pins. This makes them behave similarly to flashless RP2350 devices in the following ways:

- The QSPI **CSn** can be driven low at reset or power-up to select BOOTSEL mode
 - This harmlessly selects the internal flash die but does not issue commands to it
- The QSPI **SD1** pin can be driven high when selecting BOOTSEL to choose UART boot
 - UART TX appears on **SD2** and UART RX on **SD3**, as per [Section 5.8](#)
 - Even with the chip select asserted low, the internal flash die maintains a high-impedance state on its **SD0** through **SD3** pins if there are no transitions on **SCK**, so you can keep **CSn** asserted throughout UART boot
- Internal flash can be programmed via UF2 drag-and-drop download using the USB BOOTSEL mode
- A second QSPI device can be attached externally by connecting it to the QSPI pins and a secondary chip select from the Bank 0 GPIOs
 - This may be used for additional flash capacity, or external QSPI RAM

- See [Section 12.14](#) for more details of the RP2350 QSPI memory interface and its capabilities

The internal flash die can also be programmed externally by holding the RP2350 die in reset via the **RUN** pin (active-low reset), and driving QSPI signals *into* the chip from an external programmer.

The internal flash is powered by the QSPI_IOVDD supply input. This voltage must be in the range 2.7 to 3.6 V. You should account for the increased high-frequency currents on this supply pin in your decoupling circuit and PCB layout.

The maximum QSPI clock frequency of the W25Q16JVWI is 133 MHz. Consult the W25Q16JVWI datasheet for detailed timings and AC parameters.

If you do not require access to the RP2350 QSPI bus from the outside, you should minimise the track length connected to the QSPI package pins on your PCB. This avoids unnecessary emissions and capacitive loading of the QSPI bus.

The PADRESETB reset input on the W25Q16JVWI is not connected to any external package pins, or to any internal signals on the RP2350 die. This means there is no way to perform a hardware reset of the flash die. When the RP2350 die comes out of reset it initialises the flash die in the same way it would an external flash device by issuing a fixed XIP exit sequence that returns the flash die to a serial command state in preparation for execute-in-place setup.

14.4. Package Markings

RP2350 comes in 7 × 7 mm QFN-60 and 10 × 10 mm QFN-80 packages which are marked with the following data:

- Pin 1 Dot
- Logo
- Part number
- Date code (Week)
- Silicon lot number
- Date code (Year)

Part number includes the following:

- Device name "RP2350"
- Package type, "A" for QFN-60 or "B" for QFN-80
- Revision "0A2"

14.5. Storage conditions

In order to preserve the shelf and floor life of bare RP2350 devices, follow JEDEC J-STD (020E & 033D).

RP2350 QFN-60 is classified as Moisture Sensitivity Level 1 (MSL1). The MSL of QFN-80 is still being characterised and details will follow in a future datasheet update.

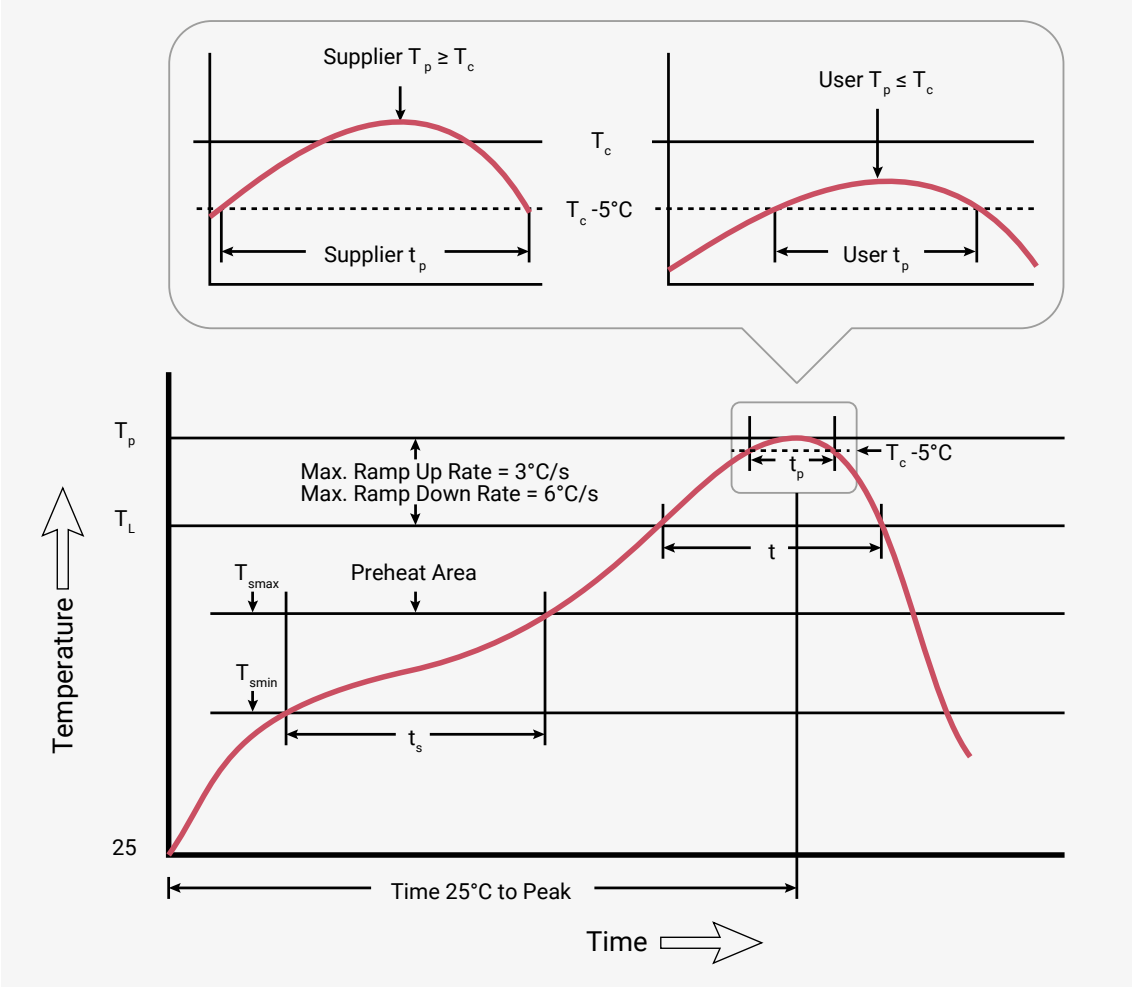
All RP2350 devices should be stored under 30°C and 85% relative humidity.

14.6. Solder profile

RP2350 is a Pb-free part, with a T_p value of 260°C.

All temperatures refer to the centre of the package, measured on the package body surface that faces up during assembly reflow (live-bug orientation). If parts are reflowed in a different orientation (e.g. dead-bug), T_p shall be within $\pm 2^\circ\text{C}$ of the live-bug T_p and still meet the T_c requirements; otherwise, you must adjust the profile to achieve the latter.

Figure 146.
Classification profile
(not to scale)



NOTE

Reflow profiles in this document are for classification/preconditioning, and are not meant to specify board assembly profiles. Actual board assembly profiles should be developed based on specific process needs and board designs, and should not exceed the parameters in [Table 1424](#).

Table 1424. Solder
profile values

Profile feature	Value
Temperature min (T_{smin})	150°C
Temperature max (T_{smax})	200°C
Time (t_s) from (T_{smin} to T_{smax})	60 - 120 seconds
Ramp-up rate (T_L to T_p)	3°C/second max.
Liquidous temperature (T_L)	217°C
Time (t_L) maintained above T_L	60 to 150 seconds
Peak package body temperature (T_p)	260°C
Classification temperature (T_c)	260°C
Time (t_p) within 5°C of the specified classification temperature (T_c)	30 seconds
Ramp-down rate (T_p to T_L)	6°C/second max.
Time 25°C to peak temperature	8 minutes max.

14.7. Compliance

RP2350 QFN-60 is compliant to Moisture Sensitivity Level 1. The Moisture Sensitivity Level compliance of RP2350 QFN-80 is yet to be fully characterised, and details will follow in a future datasheet update.

RP2350 is compliant to the requirement of REACH Substances of Very High Concern (SVHC), EU ECHA directive.

RP2350 is compliant to the requirement and standard of Controlled Environment-related Substance of RoHS directive (EU) 2011/65/EU and directive (EU) 2015/863.

Raspberry Pi Ltd carried out the following Package Level reliability qualifications on RP2350:

- Temperature Cycling per JESD22-A104
- HAST per JESD22-A110
- HTSL per JESD22-A103
- MSL level per JESD22-A113

The following Silicon Level reliability qualification were also carried out:

- HTOL per JESD22-A108F

NOTE

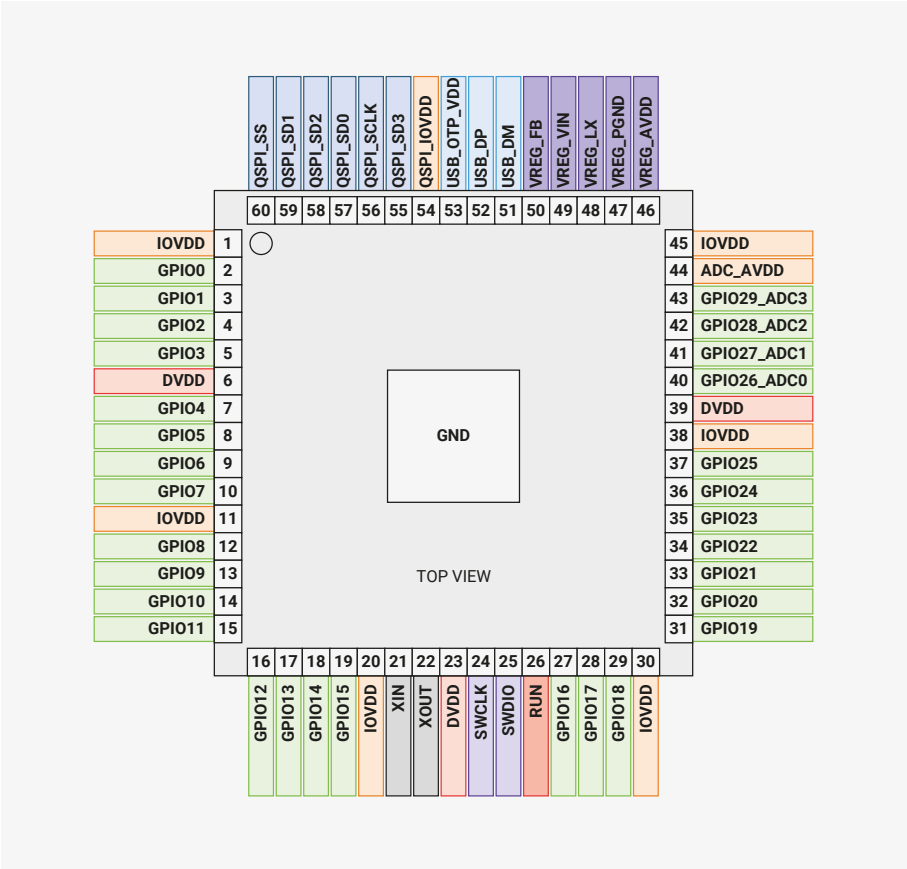
A tin whiskers test is not performed. RP2350 is a bottom-only termination device in the QFN-60 and QFN-80 packages, therefore JEDEC standard (JESD201A) is not applicable.

14.8. Pinout

14.8.1. Pin Locations

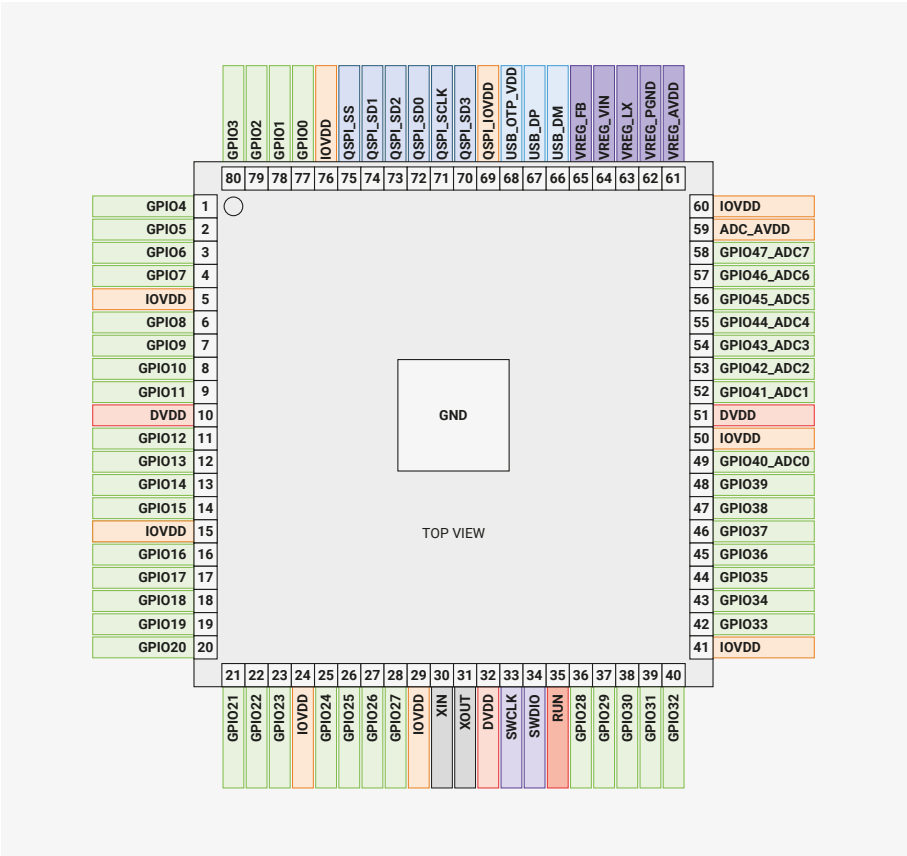
14.8.1.1. QFN-60 (RP2350A)

Figure 147. RP2350
Pinout for QFN-60
7×7mm



14.8.1.2. QFN-80 (RP2350B)

Figure 148. RP2350
Pinout for QFN-80
10×10mm



14.8.2. Pin Definitions

14.8.2.1. Pin Types

In the following pin tables (Table 1426), the pin types are defined as shown below.

Table 1425. Pin Types

Pin Type	Direction	Description
Digital In	Input only	Standard Digital. Programmable Pull-Up, Pull-Down, Slew Rate, Schmitt Trigger and Drive Strength. Default Drive Strength is 4mA.
Digital IO	Bi-directional	
Digital In (FT)	Input only	Fault Tolerant Digital. These pins are described as Fault Tolerant, which in this case means that very little current flows into the pin whilst it is below 3.63V and IOVDD is 0V. These pins have enhanced ESD protection. Programmable Pull-Up, Pull-Down, Slew Rate, Schmitt Trigger and Drive Strength. Default Drive Strength is 4mA.
Digital IO (FT)	Bi-directional	
Digital IO / Analogue	Bi-directional (digital), Input (Analogue)	Standard Digital and ADC input. Programmable Pull-Up, Pull-Down, Slew Rate, Schmitt Trigger and Drive Strength. Default Drive Strength is 4mA.
USB IO	Bi-directional	These pins are for USB use, and contain internal pull-up and pull-down resistors, as per the USB specification. USB operation requires external 27Ω series resistors.
Analogue (XOSC)		Oscillator input pins for attaching a 12MHz crystal. Alternatively, XIN may be driven by a square wave.

14.8.2.2. Pin List

Table 1426. GPIO pins

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Reset State	Description
GPIO0	2	77	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO1	3	78	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO2	4	79	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO3	5	80	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO4	7	1	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO5	8	2	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO6	9	3	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO7	10	4	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO8	12	6	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO9	13	7	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO10	14	8	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO11	15	9	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO12	16	11	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO13	17	12	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO14	18	13	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPIO15	19	14	Digital IO (FT)	IOVDD	Pull-Down	User IO

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Reset State	Description
GPI016	27	16	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI017	28	17	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI018	29	18	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI019	31	19	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI020	32	20	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI021	33	21	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI022	34	22	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI023	35	23	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI024	36	25	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI025	37	26	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI026_ADC0	40	-	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPI027_ADC1	41	-	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPI028_ADC2	42	-	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPI029_ADC3	43	-	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPI026	-	27	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI027	-	28	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI028	-	36	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI029	-	37	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI030	-	38	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI031	-	39	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI032	-	40	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI033	-	42	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI034	-	43	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI035	-	44	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI036	-	45	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI037	-	46	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI038	-	47	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI039	-	48	Digital IO (FT)	IOVDD	Pull-Down	User IO
GPI040_ADC0	-	49	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPI041_ADC1	-	52	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPI042_ADC2	-	53	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Reset State	Description
GPIO43_ADC3	-	54	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPIO44_ADC4	-	55	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPIO45_ADC5	-	56	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPIO46_ADC6	-	57	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input
GPIO47_ADC7	-	58	Digital IO / Analogue	IOVDD / ADC_AVDD	Pull-Down	User IO or ADC input

Table 1427. QSPI pins

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Reset State	Description
QSPI_SD3	55	70	Digital IO	QSPI_IOVDD	Pull-Up	QSPI data
QSPI_SCLK	56	71	Digital IO	QSPI_IOVDD	Pull-Down	QSPI clock
QSPI_SD0	57	72	Digital IO	QSPI_IOVDD	Pull-Down	QSPI data
QSPI_SD2	58	73	Digital IO	QSPI_IOVDD	Pull-Up	QSPI data
QSPI_SD1	59	74	Digital IO	QSPI_IOVDD	Pull-Down	QSPI data
QSPI_SS	60	75	Digital IO	QSPI_IOVDD	Pull-Up	QSPI chip select / USB BOOTSEL

Table 1428. Crystal oscillator pins

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Description
XIN	21	30	Analogue (XOSC)	IOVDD	Crystal oscillator. XIN may also be driven by a square wave.
XOUT	22	31	Analogue (XOSC)	IOVDD	Crystal oscillator.

Table 1429. Miscellaneous pins

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Reset State	Description
RUN	26	35	Digital In (FT)	IOVDD	Pull-Up	Chip enable / reset_n
SWCLK	24	33	Digital In (FT)	IOVDD	Pull-Up	Serial Wire Debug clock
SWDIO	25	34	Digital IO (FT)	IOVDD	Pull-Up	Serial Wire Debug data

Table 1430. USB pins

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Description
USB_DP	52	67	USB IO	USB_OTP_VDD	USB Data +ve. 27Ω series resistor required for USB operation

Name	QFN-60 Number	QFN-80 Number	Type	Power Domain	Description
USB_DM	51	66	USB IO	USB_OTP_VDD	USB Data -ve. 27Ω series resistor required for USB operation

Table 1431. Power supply pins

Name	QFN-60 Number(s)	QFN-80 Number(s)	Description
DVDD	6, 23, 39	10, 32, 51	Core supply
IOVDD	11, 20, 30, 38, 45, 54	5, 15, 24, 29, 41, 50, 60, 76	IO supply
QSPI_IOVDD	54	69	QSPI IO supply
USB_OTP_VDD	53	68	USB & OTP supply
ADC_AVDD	44	59	ADC supply
VREG_AVDD	46	61	Voltage regulator analogue supply
VREG_PGND	47	62	Voltage regulator ground
VREG_LX	48	63	Voltage regulator switching output (connect to inductor)
VREG_VIN	49	64	Voltage regulator input supply
VREG_FB	50	65	Voltage regulator feedback input
GND	-	-	Ground connection via central exposed pad

14.9. Electrical Specifications

The following electrical specifications are obtained from characterisation over the specified temperature and voltage ranges, as well as process variation, unless the specification is marked as 'Simulated'. In this case, the data is for information purposes only, and is not guaranteed.

14.9.1. Absolute Maximum Ratings

Table 1432. Absolute maximum ratings

Parameter	Symbol	Conditions	Minimum	Maximum	Units	Comment
Core Supply (DVDD) Voltage	DVDD		-0.5	1.21	V	
I/O Supply (IOVDD) & QSPI Supply (QSPI_IOVDD) Voltage	IOVDD		-0.5	3.63	V	
Voltage at IO (Standard)	V _{PIN}		-0.5	IOVDD + 0.5	V	

Parameter	Symbol	Conditions	Minimum	Maximum	Units	Comment
Voltage at IO (FT)	V_{PIN_FT}	IOVDD=3.3V	-0.5	5.5	V	IOVDD must be present
		IOVDD=2.5V	-0.5	4.2	V	
		IOVDD=1.8V	-0.5	3.63	V	
		IOVDD=0V	-0.5	3.63	V	

14.9.2. ESD Performance

Table 1433. ESD performance for all pins, unless otherwise stated

Parameter	Symbol	Maximum	Units	Comment
Human Body Model	HBM	2	kV	Compliant with JEDEC specification JS-001-2012 (April 2012)
Human Body Model Digital (FT) pins only	HBM	4	kV	Compliant with JEDEC specification JS-001-2012 (April 2012)
Charged Device Model	CDM	500	V	Compliant with JESD22-C101E (December 2009)

14.9.3. Thermal Performance

Table 1434. Thermal Performance

Parameter	Symbol	Minimum	Typical	Maximum	Units	Comment
Case Temperature	T_C	-40		85	°C	

14.9.4. IO Electrical Characteristics

Table 1435. Digital IO characteristics - Standard and FT unless otherwise stated. In this table IOVDD also refers to QSPI_IOVDD where appropriate

Parameter	Symbol	Conditions	Minimum	Maximum	Units	Comment
Pin Input Leakage Current	I_{IN}			1	μA	
Input Voltage High (Standard IO)	V_{IH}	IOVDD=1.8V	0.65 * IOVDD	IOVDD + 0.3	V	
		IOVDD=2.5V	1.7	IOVDD + 0.3	V	
		IOVDD=3.3V	2	IOVDD + 0.3	V	
Input Voltage High (FT)	V_{IH}	IOVDD=1.8V	0.65 * IOVDD	3.63	V	IOVDD must be powered to tolerate input voltages above 3.63V
		IOVDD=2.5V	1.7	4.2	V	
		IOVDD=3.3V	2	5.5	V	
Input Voltage Low	V_{IL}	IOVDD=1.8V	-0.3	0.35 * IOVDD	V	
		IOVDD=2.5V	-0.3	0.7	V	
		IOVDD=3.3V	-0.3	0.8	V	

Parameter	Symbol	Conditions	Minimum	Maximum	Units	Comment
Input Hysteresis Voltage	V_{HYS}	IOVDD=1.8V	$0.1 * IOVDD$		V	Schmitt Trigger enabled
		IOVDD=2.5V	0.2		V	
		IOVDD=3.3V	0.2		V	
Output Voltage High	V_{OH}	IOVDD=1.8V	1.24	IOVDD	V	$I_{OH} = 2, 4, 8$ or $12mA$ depending on setting
		IOVDD=2.5V	1.78	IOVDD	V	
		IOVDD=3.3V	2.62	IOVDD	V	
Output Voltage Low	V_{OL}	IOVDD=1.8V	0	0.3	V	$I_{OL} = 2, 4, 8$ or $12mA$ depending on setting
		IOVDD=2.5V	0	0.4	V	
		IOVDD=3.3V	0	0.5	V	
Pull-Up Resistance	R_{PU}	IOVDD=1.8V	32	106	k Ω	
		IOVDD=2.5V	42	123	k Ω	
		IOVDD=3.3V	32	86	k Ω	
Pull-Down Resistance	R_{PD}	IOVDD=1.8V	35	189	k Ω	
		IOVDD=2.5V	49	180	k Ω	
		IOVDD=3.3V	36	113	k Ω	
Maximum Total IOVDD current	I_{IOVDD_MAX}			100	mA	Sum of all current being sourced by GPIO pins
Maximum Total QSPI_IOVDD current	$I_{QSPI_IOVDD_MAX}$			20	mA	Sum of all current being sourced by QSPI pins
Maximum Total VSS current due to GPIO (IOVSS)	I_{IOVSS_MAX}			100	mA	Sum of all current being sunk into GPIO pins
Maximum Total VSS current due to QSPI (QSPI_IOVSS)	$I_{QSPI_IOVSS_MAX}$			20	mA	Sum of all current being sunk into QSPI pins

Table 1436. USB IO characteristics

Parameter	Symbol	Minimum	Maximum	Units	Comment
Pin Input Leakage Current	I_{IN}		1	μA	
Single Ended Input Voltage High	V_{IHSE}	2		V	
Single Ended Input Voltage Low	V_{ILSE}		0.8	V	
Differential Input Voltage High	V_{IHDIFF}	0.2		V	

Parameter	Symbol	Minimum	Maximum	Units	Comment
Differential Input Voltage Low	V_{ILDIFF}		-0.2	V	
Output Voltage High	V_{OH}	2.8	USB_OTG_VDD	V	
Output Voltage Low	V_{OL}	0	0.3	V	
Pull-Up Resistance - RPU2	R_{PU2}	0.873	1.548	k Ω	
Pull-Up Resistance - RPU1&2	$R_{PU1\&2}$	1.398	3.063	k Ω	
Pull-Down Resistance	R_{PD}	14.25	15.75	k Ω	

Table 1437. ADC characteristics

Parameter	Symbol	Minimum	Typical	Maximum	Units	Comment
ADC Input Voltage Range	V_{PIN_ADC}	0		ADC_AVDD	V	
Effective Number of Bits	ENOB	9	9.5		bits	
Resolved Bits				12	bits	
ADC Input Impedance	R_{IN_ADC}	100			k Ω	

Table 1438. Oscillator pin characteristics

Parameter	Symbol	Minimum	Typical	Maximum	Units	Comment
Input Frequency	f_{osc}	1	12	50	MHz	See Section 8.6.3 for restrictions imposed by PLLs. See Section 5.2.8.1 for restrictions imposed by the USB and UART bootloaders.
Input Voltage High	V_{IH}	0.65*IOVDD		IOVDD + 0.3	V	Square Wave input. XIN only. XOUT floating
Input Voltage Low	V_{IL}	0		0.35*IOVDD	V	Square Wave input. XIN only. XOUT floating

NOTE

By default, USB Bootmode relies on a 12MHz input being present. However OTP can be configured to override the XOSC and PLL settings during USB Bootmode. See [Section 13.9](#) for details.

Table 1439. SWCLK pin characteristics

Parameter	Symbol	Minimum	Typical	Maximum	Units	Comment
SWCLK Input Frequency	f_{SWCLK}	0	10	50	MHz	See Table 1429 for SWCLK pin definitions.

Host-to-target data on the SWDIO pin should be transmitted centre-aligned with SWCLK. Target-to-host data on the SWDIO pin transitions on rising edges of SWCLK.

NOTE

RP2350 internal SWD logic in the SW-DP operates reliably up to 50 MHz. However, signal integrity of the external SWD signals may be a challenge.

If you observe unreliable SWD operation such as write data parity errors from the SW-DP, reduce the SWCLK frequency. Always connect ground directly between the SWD probe and RP2350 in addition to SWDIO and SWCLK. Minimise the wire length between the probe and RP2350, and avoid multi-drop wiring at higher frequencies.

14.9.4.1. Interpreting GPIO output voltage specifications

The GPIOs on RP2350 have four different output drive strengths, nominally called 2, 4, 8 and 12mA modes. These are not hard limits, nor do they mean that they will always source (or sink) the selected amount of milliamps.

The amount of current a GPIO sources or sinks is dependent on the load attached. It will attempt to drive the output to the IOVDD level (or 0V in the case of a logic 0), but the amount of current it is able to source is limited and dependent on the selected drive strength.

Therefore the higher the current load is, the lower the voltage will be at the pin. At some point, the GPIO will source so much current and the voltage will drop so low that it won't be recognised as a logic 1 by the input of a connected device. The output specifications in [Table 1435](#) quantify how much lower the voltage can be expected to be when drawing specified amounts of current from the pin.

The Output High Voltage (V_{OH}) is defined as the lowest voltage the output pin can be when driven to a logic 1 with a particular selected drive strength; e.g., 4mA sourced by the pin whilst in 4mA drive strength mode. The Output Low Voltage is similar, but with a logic 0 being driven.

In addition to this, the sum of all the IO currents being sourced (i.e. when outputs are being driven high) from the IOVDD bank (essentially the GPIO and QSPI pins), must not exceed I_{IOVDD_MAX} . Similarly, the sum of all the IO currents being sunk (i.e. when the outputs are being driven low) must not exceed I_{IOVSS_MAX} .

Figure 149. Typical Current vs Voltage curves of a GPIO output.

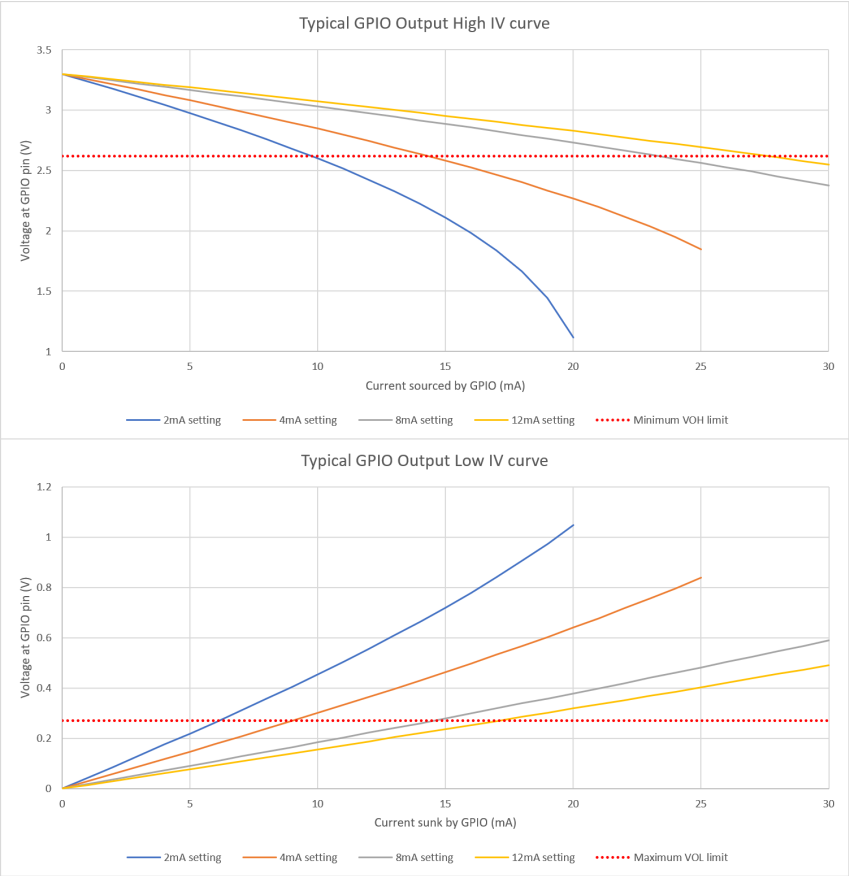


Figure 149 shows the effect on the output voltage as the current load on the pin increases. You can clearly see the effect of the different drive strengths; the higher the drive strength, the closer the output voltage is to IOVDD (or 0V) for a given current. The minimum V_{OH} and maximum V_{OL} limits are shown in red.

You can see that at the specified current for each drive strength, the voltage is well within the allowed limits, meaning that this particular device could drive a lot more current and still be within V_{OH}/V_{OL} specification. This is a typical part at room temperature, but because devices vary, there will be a spread of other devices which will have voltages much closer to this limit.

If your application doesn't need such tightly controlled voltages, you can source or sink more current from the GPIO than the selected drive strength setting. However, experimentation is required to determine if it indeed safe to do so in your application.

14.9.5. Power Supplies

Table 1440. Power Supply Specifications

Power Supply	Supplies	Min	Typ	Max	Units
IOVDD ^a	Digital IO	1.62	1.8 / 3.3	3.63	V
QSPI_IOVDD (RP2350 only) ^a	Digital IO	1.62	1.8 / 3.3	3.63	V
QSPI_IOVDD (RP2354 only)	Digital IO	2.97	3.3	3.63	V
DVDD ^b	Digital core	1.05	1.1	1.16	V
VREG_VIN	Voltage regulator	2.7	3.3	5.5	V
VREG_AVDD	Voltage regulator	3.135	3.3	3.63	V

Power Supply	Supplies	Min	Typ	Max	Units
USB_OTP_VDD	USB PHY & OTP	3.135	3.3	3.63	V
ADC_AVDD ^c	ADC	1.62	3.3	3.63	V

^a If IOVDD < 2.5V, GPIO `VOLTAGE_SELECT` registers should be adjusted accordingly. See [Section 6.1](#) for details.

^b Short term transients should be within +/-100mV.

^c ADC performance will be compromised at voltages below 2.97V

i NOTE

RP2354 contains an internal 3.3V flash device, therefore `QSPI_IOVDD` must be 3.3V. Furthermore, if the QSPI pins are to be used to connect to an additional flash or PSRAM device, then IOVDD must be 3.3V, as a GPIO is used as QSPI chip select in this case.

14.9.6. Core Voltage Regulator

Table 1441. Voltage Regulator Specifications

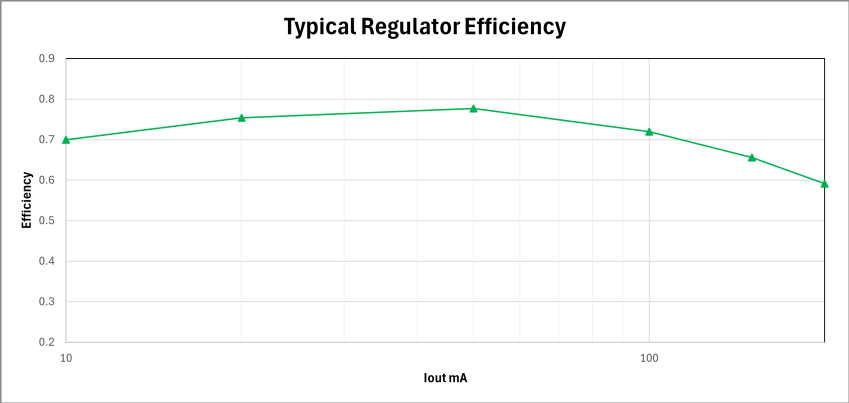
Parameter	Description	Min	Typ	Max	Units
V _{OUT} (normal mode)	regulated output voltage range (normal mode)	0.55	1.1	3.3	V
V _{OUT} (low power mode)	regulated output voltage range (low power mode)	0.55	1.1	1.3	V
ΔV _{OUT} (normal mode)	voltage deviation from programmed value (normal mode)	-3		+3	% of selected output voltage
ΔV _{OUT} (low power mode)	voltage deviation from programmed value (low power mode)	-9		+9	% of selected output voltage
I _{MAX} (normal mode)	output current (normal mode)			200	mA
I _{MAX} (low power mode)	output current (low power mode)			1	mA
I _{LIMIT} (normal mode startup)	current limit (normal mode startup)		240	300	mA
I _{LIMIT} (normal mode)	current limit (normal mode)	260	500	800	mA
I _{LIMIT} (low power mode)	current limit (low power mode)	5		25	mA
VOUT_OK _{TH.ASSERT}	<code>VOUT_OK</code> assertion threshold	87	90	93	% of selected output voltage

Parameter	Description	Min	Typ	Max	Units
VOUT_OK _{TH.DEASSERT}	VOUT_OK de-assertion threshold	84	87	90	% of selected output voltage
f _{sw}	switching frequency		3		MHz
Efficiency (V _{OUT} =1.1V)	I _{load} =10mA, VREG_VIN=2.7V		74		%
	I _{load} =10mA, VREG_VIN=3.3V		70		%
	I _{load} =10mA, VREG_VIN=5.5V		59		%
	I _{load} =100mA, VREG_VIN=2.7V		70		%
	I _{load} =100mA, VREG_VIN=3.3V		72		%
	I _{load} =100mA, VREG_VIN=5.5V		72		%
	I _{load} =200mA, VREG_VIN=2.7V		70		%
	I _{load} =200mA, VREG_VIN=3.3V		59		%
	I _{load} =200mA, VREG_VIN=5.5V		63		%

WARNING

V_{OUT} can exceed the maximum core supply (DVDD). While there is a voltage limit to prevent this happening accidentally, the limit can be disabled under software control. For reliable operation DVDD should not exceed its maximum voltage rating.

Figure 150. Typical Regulator Efficiency, V_{OUT}=1.1V, VREG_VIN=3.3V.



14.9.7. Power Consumption

14.9.7.1. Peripheral power consumption

Baseline readings are taken with only clock sources and essential peripherals (BUSCTRL, BUSFAB, VREG, Resets, ROM, SRAMs) active in the `WAKE_EN0/WAKE_EN1` registers. Clocks are set to default clock settings.

Each peripheral is activated in turn by enabling all clock sources for the peripheral in the `WAKE_EN0/WAKE_EN1` registers. Current consumption is the increase in current when the peripheral clocks are enabled.

Table 1442. Baseline power consumption

Peripheral	Typical DVDD Current Consumption (µA/MHz)
DMA	2.6
I2C0	3
I2C1	3.6
IO + Pads	24.5
PWM	9.9
SIO	2
SHA256	0.1
SPI0	1.7
SPI1	1.4
Timer 0	0.8
Timer 1	0.6
TRNG	0.8
UART0	2.6
UART1	3.6
Watchdog	1.1
XIP	37.6

Because of fixed reference clocks of 48MHz, as well as the variable system clock input, ADC and USBCTRL power consumption does not vary linearly with system clock (as it does for other peripherals which only have system and/or peripheral clock inputs). The following table shows absolute DVDD current consumption of the ADC and USBCTRL blocks at standard clocks settings:

Table 1443. Baseline power consumption for ADC and USBCTRL

Peripheral	Typical DVDD Current Consumption (mA)
ADC	0.14
USBCTRL	1.25

14.9.7.2. Power consumption in Low Power states

Table 1444 shows the typical power consumption in low power states `P1.0` → `P1.7`. All voltage supplies are 3.3V (except DVDD which is supplied by the voltage regulator (in low power mode)), with the environment at room temperature.

All GPIOs, `SWDIO` and `SWCLK` are pulled down internally, and not connected externally. QSPI is connected to W25Q16JVSSIQ flash device. USB PHY has been powered down, and the `DP` and `DM` pull-downs were enabled prior to entering the low power state. The USB cable remains connected to a host computer. The table also shows the power consumed when `RUN` is held low. This is not technically a low power state (the voltage regulator is in normal switching mode), but it is included for completeness.

Table 1444. Low Power States Power Consumption

Low Power State	VREG_VIN (μA)	VREG_AVDD (μA)	IOVDD (μA)	QSPI_IOVDD (μA)	ADC_IOVDD (μA)	USB_OTP_VDD (μA)	Total Power (μW)
P1.0	128	0.5	11	22	1	3.5	548
P1.1	77	0.5	11	22	1	3.5	380
P1.2	79	0.5	11	22	1	3.5	380
P1.3	26	0.5	11	22	1	3.5	204
P1.4	120	0.5	11	22	1	3.5	520
P1.5	67	0.5	11	22	1	3.5	345
P1.6	68	0.5	11	22	1	3.5	350
P1.7	19	0.5	11	22	1	3.5	188
RUN=low	40	187	69	22	1	35	1170

14.9.7.3. Power consumption for typical user cases

The following table details the typical power consumption of RP2350 in various example use cases. All measurements were taken using 3.3V voltage supplies (except DVDD, which is supplied by the voltage regulator (set to 1.1V)), with the environment at room temperature.

SWD and **SWCLK** are not connected externally. **GPIO0** and **GPIO1** are connected to a Raspberry Pi Debug Probe (UART), but all other GPIOs are not connected (except USB Boot mode, where **GPIO0** and **GPIO1** are also unconnected). **QSPI** is connected to W25Q16JVSSIQ flash device, and **USB** is connected to a host.

hello_serial, **hello_usb** and **hello_adc** are basic applications found in pico-examples where characters are constantly transmitted to a serial console.

Table 1445. Power Consumption

Use Case	Condition	VREG_VIN (μA)	VREG_AVDD (μA)	IOVDD (μA)	QSPI_IOVDD (μA)	ADC_IOVDD (μA)	USB_OTP_VDD (μA)	Total Power (mW)
USB Boot mode	Bus Idle (average)	6530	220	437	22	1	375	25
	During Boot (peak)						6050	
	During UF2 write (average)						1280	
hello_serial		14690	216	506	22	1	62	51.1
hello_usb		14700	216	453	22	1	570	52.7
hello_adc		14680	216	506	22	142	62	51.6
CoreMark benchmark	Single core @150MHz	11000	212	455	22	1	90	38.7

14.9.7.3.1. Power Consumption versus frequency

There is a relationship between the core RP2350 frequency and the current consumed by the DVDD supply. [Figure 151](#) shows the measured results of a typical RP2350 device that continuously runs CoreMark benchmark tests on a single core at various core clock frequencies.

Figure 151. DVDD
Current vs Core
Frequency of a typical
RP2350 device, whilst
running CoreMark
benchmark

